

CS5340 - Uncertainty Modelling in AI

Lecture 3: Markov Random Fields (Undirected Graphical Models)

Scribed by: Claude (Anthropic) | Curated By: Assoc. Prof. Lee Gim Hee

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1 Lecture Overview

- **Motivation:** Bayesian Networks (directed graphical models) require us to choose an orientation for every edge – for some domains (e.g. spatial or pixel data) this requirement is **awkward** and unnatural.
- **Goal:** introduce **Markov Random Fields (MRFs)**, a.k.a. **Undirected Graphical Models (UGMs)**, as an alternative, edge-orientation-free representation of conditional independence.
- **Roadmap:**
 1. Derive five additional rules of conditional independence.
 2. Use the **Independence-Map (I-Map)** to check whether a graph validly represents a distribution.
 3. Explain the **Markov properties** (global, local, pairwise) of a UGM and use them to read off all conditional independencies.
 4. **Parameterize** an MRF using clique potential functions (Hammersley–Clifford theorem).
 5. Connect MRFs to **Gibbs distributions** and **log-linear** potentials.
 6. Compare MRFs with **Conditional Random Fields (CRFs)**.

Learning Outcomes

By the end of this lecture, you should be able to:

1. Derive the five rules of conditional independence: Symmetry, Decomposition, Weak Union, Contraction, Intersection.
2. Use I-Map to explain the correctness of graphical models for conditional independence.
3. Explain the concepts of Markov properties (global, local and pairwise) and use them to find all conditional independencies in a UGM.
4. Use clique potential functions to parameterize a Markov Random Field, i.e. represent the joint distribution with clique potential functions.
5. Describe the differences and similarities between a Markov Random Field and a Conditional Random Field.

2 Rules of Conditional Independence

Recall from the last lecture that conditional independence $X_A \perp X_C \mid X_B$ means

$$X_A \perp X_C \mid X_B \implies \begin{cases} p(x_A, x_C \mid x_B) = p(x_A \mid x_B) p(x_C \mid x_B) \\ p(x_A \mid x_B, x_C) = p(x_A \mid x_B), \end{cases} \quad \forall x_B : p(x_B) > 0.$$

2.1 Statement of the Five Rules

Each rule below is stated as a **one-directional** implication – whether its converse also holds is a separate question, addressed in Section 2.3 once all five rules have been proved.

Key Rule: Five Rules of Conditional Independence

1. **Symmetry:** $X \perp Y \mid Z \implies Y \perp X \mid Z$.
2. **Decomposition:** $X \perp (Y, W) \mid Z \implies X \perp Y \mid Z$ and $X \perp W \mid Z$.
3. **Weak Union:** $X \perp (Y, W) \mid Z \implies X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z)$.
4. **Contraction:** $X \perp Y \mid (W, Z)$ and $X \perp W \mid Z \implies X \perp (Y, W) \mid Z$.
5. **Intersection (requires $p(x) > 0$):** $X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z) \implies X \perp (Y, W) \mid Z$.

Positivity: only **Intersection** needs the strict-positivity assumption $p(x) > 0$ for all x . Rules 1–4 hold for **any** probability distribution.

2.2 Proofs of the Five Rules

Proof: Symmetry

Using the product rule, $p(Y \mid X, Z) = \frac{p(X, Y \mid Z)}{p(X \mid Z)}$, $\forall p(X \mid Z) > 0$.

Applying $X \perp Y \mid Z$ to $p(X, Y \mid Z)$:

$$p(Y \mid X, Z) = \frac{p(X, Y \mid Z)}{p(X \mid Z)} = \frac{p(X \mid Z) p(Y \mid Z)}{p(X \mid Z)} = p(Y \mid Z).$$

Thus Y is conditionally independent of X given Z , i.e. $Y \perp X \mid Z$. ■

Proof: Decomposition

$$X \perp \{Y, W\} \mid Z \implies p(X, Y, W \mid Z) = p(X \mid Z) p(Y, W \mid Z).$$

$$\begin{aligned} p(X, Y \mid Z) &= \int p(X, Y, W \mid Z) dW = \int p(X \mid Z) p(Y, W \mid Z) dW = p(X \mid Z) p(Y \mid Z) \\ &\implies X \perp Y \mid Z. \end{aligned}$$

$X \perp W \mid Z$ can be shown in a similar way (marginalize over Y instead of W). ■

Proof: Weak Union

$$\begin{aligned}
p(X, Y \mid W, Z) &= \frac{p(X, Y, W \mid Z)}{p(W \mid Z)} && \forall p(W \mid Z) > 0 \text{ (product rule)} \\
&= \frac{p(X \mid Z) p(Y, W \mid Z)}{p(W \mid Z)} && \text{since } X \perp \{Y, W\} \mid Z \\
&= p(X \mid Z) p(Y \mid W, Z) && \text{(product rule)} \\
&= p(X \mid W, Z) p(Y \mid W, Z) && \text{Decomposition: } X \perp \{Y, W\} \mid Z \Rightarrow X \perp W \mid Z
\end{aligned}$$

Therefore $X \perp \{Y, W\} \mid Z \Rightarrow X \perp Y \mid (W, Z)$; the statement $X \perp W \mid (Y, Z)$ is proven the same way (swap the roles of Y and W throughout). ■

Proof: Contraction

By hypothesis $X \perp Y \mid (W, Z)$, so $p(X \mid Y, W, Z) = p(X \mid W, Z)$. By hypothesis $X \perp W \mid Z$, so $p(X \mid W, Z) = p(X \mid Z)$. Chaining these two equalities:

$$p(X \mid Y, W, Z) = p(X \mid W, Z) = p(X \mid Z).$$

Multiplying through by $p(Y, W \mid Z)$ and using the product rule $p(X, Y, W \mid Z) = p(X \mid Y, W, Z) p(Y, W \mid Z)$:

$$\begin{aligned}
p(X, Y, W \mid Z) &= p(X \mid Y, W, Z) p(Y, W \mid Z) = p(X \mid Z) p(Y, W \mid Z) \\
&\Rightarrow X \perp \{Y, W\} \mid Z.
\end{aligned}$$

■

Proof: Intersection (requires $p(x) > 0$)

$$\begin{aligned}
X \perp Y \mid \{W, Z\} &\Rightarrow p(X \mid W, Y, Z) = p(X \mid W, Z), \\
X \perp W \mid \{Y, Z\} &\Rightarrow p(X \mid W, Y, Z) = p(X \mid Y, Z) \\
&\Rightarrow p(X \mid W, Z) = p(X \mid Y, Z).
\end{aligned}$$

Continuing (marginalizing over W , then applying the product rule and the two hypotheses):

$$\begin{aligned}
p(X \mid Z) &= \int p(X, W \mid Z) dW && \text{(marginalization)} \\
&= \int p(X \mid W, Z) p(W \mid Z) dW && \text{(product rule)} \\
&= p(X \mid Y, Z) \int p(W \mid Z) dW && X \perp Y \mid \{W, Z\}, X \perp W \mid \{Y, Z\} \\
&= p(X \mid Y, Z)
\end{aligned}$$

$$\Rightarrow p(X \mid W, Y, Z) = p(X \mid W, Z) = p(X \mid Y, Z) = p(X \mid Z) \Rightarrow X \perp \{Y, W\} \mid Z. \blacksquare$$

Where positivity is used: the equality $p(X \mid W, Z) = p(X \mid Y, Z)$ above was obtained by chaining $p(X \mid W, Y, Z) = p(X \mid W, Z)$ with $p(X \mid W, Y, Z) = p(X \mid Y, Z)$ – but each of these

only holds at (w, y, z) triples with $p(w, y, z) > 0$. The next step then treats $p(X \mid W, Z) = p(X \mid Y, Z)$ as holding for *every* W with $p(W \mid Z) > 0$ (for a fixed $Y = y$), so that $p(X \mid Y, Z)$ can be pulled outside the integral over W . This is only guaranteed if every such (w, y, z) triple itself has positive joint probability – i.e. if $p(x) > 0$ everywhere. If some (w, y, z) triple has zero probability (as happens in the XOR-type example in Section 2.3), the chaining step is not licensed, the proof breaks down, and the conclusion can genuinely fail.

2.3 Do the Converses Hold?

Each rule above was proved as a **one-directional** implication. A natural question: does the **converse** (i.e. the reverse implication, making the rule an *iff*) also hold? The answer differs rule by rule – two converses hold unconditionally, one holds only under positivity, and one is **false outright**, even under positivity.

Key Rule: Which Converses Hold?

- **Symmetry** is trivially an **iff** – the same proof runs in both directions, so there is nothing further to show.
- **Decomposition**'s converse is **false in general**, with or without positivity:

$$X \perp Y \mid Z \text{ and } X \perp W \mid Z \not\Rightarrow X \perp (Y, W) \mid Z.$$

(XOR counterexample below.)

- **Weak Union**'s converse is exactly the forward direction of **Intersection**, which **requires** $p(x) > 0$. So

$$X \perp (Y, W) \mid Z \iff X \perp Y \mid (W, Z) \text{ and } X \perp W \mid (Y, Z)$$

is a genuine biconditional **only under positivity**; without it, only the forward direction (Weak Union) is guaranteed.

- **Contraction**'s converse **holds unconditionally** – it is just **Weak Union** + **Decomposition** applied together, neither of which needs positivity (proof below), giving

$$X \perp (Y, W) \mid Z \iff X \perp Y \mid (W, Z) \text{ and } X \perp W \mid Z.$$

- **Intersection**'s converse **also holds unconditionally** – it is just **Weak Union** applied twice (proof below), i.e.

$$X \perp (Y, W) \mid Z \implies X \perp Y \mid (W, Z) \text{ and } X \perp W \mid (Y, Z),$$

with **no** positivity assumption needed – positivity is only required to go *up* from the two pairwise statements to the joint one (the forward direction of Intersection), never to come back *down*.

Proof: Contraction's Converse (unconditional)

Assume $X \perp \{Y, W\} \mid Z$, i.e. $p(X \mid Y, W, Z) = p(X \mid Z)$.

$$\begin{aligned}
 p(X \mid W, Z) &= \frac{\int p(X, Y, W \mid Z) dY}{p(W \mid Z)} && \text{(product rule)} \\
 &= \frac{\int p(X \mid Y, W, Z) p(Y, W \mid Z) dY}{p(W \mid Z)} && \text{(product rule)} \\
 &= \frac{p(X \mid Z) \int p(Y, W \mid Z) dY}{p(W \mid Z)} && \text{since } p(X \mid Y, W, Z) = p(X \mid Z) \\
 &= \frac{p(X \mid Z) p(W \mid Z)}{p(W \mid Z)} = p(X \mid Z) && \text{(marginalization)} \\
 &\implies p(X \mid Y, W, Z) = p(X \mid W, Z) = p(X \mid Z)
 \end{aligned}$$

Therefore, $X \perp \{Y, W\} \mid Z \implies X \perp Y \mid (W, Z)$ and $X \perp W \mid Z$. This is really just **Weak Union** (giving $X \perp Y \mid (W, Z)$) combined with **Decomposition** (giving $X \perp W \mid Z$) – since neither needs positivity, this direction of Contraction's biconditional does not either. ■

Proof: Intersection's Converse (unconditional)

Claim: $X \perp (Y, W) \mid Z \implies X \perp Y \mid (W, Z)$ and $X \perp W \mid (Y, Z)$.

This is nothing but **Weak Union** applied twice. Applying Weak Union directly to $X \perp (Y, W) \mid Z$ gives $X \perp Y \mid (W, Z)$. Since (Y, W) is simply an unordered pair, $X \perp (Y, W) \mid Z$ is the same statement as $X \perp (W, Y) \mid Z$; applying Weak Union to that (i.e. swapping the roles of Y and W) gives $X \perp W \mid (Y, Z)$. ■

Worked Example: Decomposition's Converse Is False (XOR Counter-Example)

Let $Z = \emptyset$ for simplicity (an equivalent, simplified claim to disprove): does $X \perp Y$ and $X \perp W$ imply $X \perp \{Y, W\}$? Let $Y, W \sim \text{Bernoulli}(0.5)$ independently, and let $X = Y \oplus W$ (XOR):

Y	W	X	$P(Y, W, X)$
0	0	0	0.25
0	0	1	0
0	1	0	0
0	1	1	0.25
1	0	0	0
1	0	1	0.25
1	1	0	0.25
1	1	1	0

Marginally, $X \perp Y$ since $p(X \mid Y) = p(X) = 0.5$ for every value (and likewise $X \perp W$ since $p(X \mid W) = p(X) = 0.5$ for every value) – both pairwise independencies hold. However, jointly,

$$X \not\perp \{Y, W\} \quad \text{since} \quad p(X \mid Y, W) \in \{0, 1\} \neq p(X) = 0.5,$$

i.e. knowing *both* Y and W completely determines X (since $X = Y \oplus W$), even though knowing *either one alone* tells us nothing about X . Therefore

$$X \perp Y \text{ and } X \perp W \not\implies X \perp \{Y, W\}. \blacksquare$$

Remark: this table is **not** strictly positive ($p(y, w, x) = 0$ for half the rows), so it cannot be used to test the positivity-dependent Intersection rule – it only disproves Decomposition’s converse, which fails with or without positivity.

Summary: Forward Direction vs. Converse

Rule	Forward direction	Converse
1. Symmetry	holds (any p)	holds – identical statement
2. Decomposition	holds (any p)	fails , even if $p(x) > 0$ (XOR example)
3. Weak Union	holds (any p)	= Intersection; needs $p(x) > 0$
4. Contraction	holds (any p)	holds (any p) – Weak Union + Decomposition
5. Intersection	holds, needs $p(x) > 0$	holds (any p) – Weak Union applied twice

Take-away: rules 3, 4, and 5 are secretly two overlapping biconditionals in disguise – Contraction is an unconditional *iff*, and Weak Union/Intersection form an *iff* only under positivity. Positivity is doing real work in exactly **one** place: the forward direction of Intersection (equivalently, the converse of Weak Union). Decomposition is the genuine outlier – its converse fails regardless of any distributional assumption, because a third variable (W in the XOR example) can carry dependence on X that is invisible in Y or W alone but fully exposed once they are considered jointly.

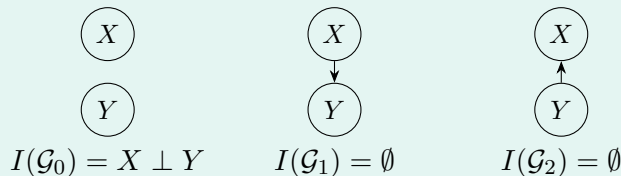
3 Independence-Map (I-Map)

Definition : Independence-Map (I-Map)

Also known as the **I-Map**.

- The I-Map of a **joint distribution** $p(x_1, \dots, x_N)$, written $I(p)$, represents **all independencies** true in $p(x_1, \dots, x_N)$.
- Similarly, the I-Map of a **directed/undirected graph** $\mathcal{G}$, i.e. $I(\mathcal{G})$, represents **all independencies** encoded in $\mathcal{G}$.
- $\mathcal{G}$ is a **valid representation** of p if $I(\mathcal{G}) \subseteq I(p)$.

Worked Example: Are $\mathcal{G}_0, \mathcal{G}_1, \mathcal{G}_2$ Valid Representations of $p(X, Y)$?



Given the joint table for $p(X, Y)$:

X	Y	$p(X, Y)$
0	0	0.08
0	1	0.32
1	0	0.12
1	1	0.48

Check for mutual independence: $p(X, Y) = p(X)p(Y)$ if $p(Y | X) = p(Y)$ and $p(X | Y) = p(X)$. Computing the marginals, $p(X=0)=0.4$, $p(X=1)=0.6$, $p(Y=0)=0.2$, $p(Y=1)=0.8$, and

indeed $p(X | Y) = p(X)$ and $p(Y | X) = p(Y)$ for every value – there **is** mutual independence, i.e. $p(X, Y) = p(X)p(Y)$.

Since $I(\mathcal{G}_0) = \{X \perp Y\} \subseteq I(p)$ and $I(\mathcal{G}_1) = I(\mathcal{G}_2) = \emptyset \subseteq I(p)$ trivially, **all three graphs** $\mathcal{G}_0, \mathcal{G}_1, \mathcal{G}_2$ are valid I-Maps of $p(X, Y)$ – a graph with *fewer* edges is not automatically a better (or worse) I-Map, as long as its encoded independencies are a subset of the true ones.

Exercise: for the joint table $p(0,0)=0.4, p(0,1)=0.3, p(1,0)=0.2, p(1,1)=0.1$, show that $\mathcal{G}_0$ is **not** an I-Map of $p(X, Y)$, while $\mathcal{G}_1$ and $\mathcal{G}_2$ still are.

Key Rule: From I-Map to Factorization

If $\mathcal{G}$ is a valid representation of P , then

$$p(X_1, \dots, X_N) = \prod_{c \in \{1, \dots, N\}} f_c(X_c),$$

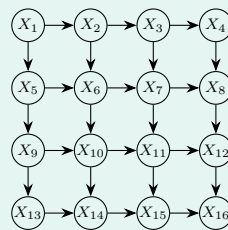
where $\prod_c f_c(X_c)$ is the **factorization** according to $\mathcal{G}$. For example, the parent-child relationship in Bayesian networks gives $p(x_1, \dots, x_N) = \prod_{i=1}^N p(x_i | x_{\pi_i})$. Thus, we can go from **graphs to factorizations** of P – this is the key idea we now develop for undirected graphs.

4 Why Undirected Graphical Models?

We discussed Directed Graphical Models (DGMs) / Bayesian Networks in the last lecture. However, for some domains the requirement for a **directed** edge is rather awkward.

Worked Example: Modeling a 2D Image (Causal MRF)

Consider modeling a 2D image where the intensity of neighboring pixels is correlated. We could force a topological ordering and create a DAG with a 2D lattice topology, known as a **causal MRF** or **Markov mesh**:



The awkward part: to model this as a DAG we must impose an arbitrary raster-scan ordering on the pixels, even though the underlying correlation structure (a neighbor is a neighbor in every direction) has no natural direction.

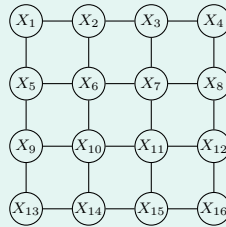
Definition : Undirected Graphical Model (UGM) / Markov Random Field (MRF)

An alternative is to use an **Undirected Graphical Model (UGM)**, also called a **Markov Random Field (MRF)** or **Markov network**.

- Formally, a UGM is a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is a set of **nodes** in one-to-one correspondence with a set of random variables, and $\mathcal{E}$ is a set of **undirected** edges.
- No edge orientations are required, hence **more natural** for problems such as **image analysis** and **spatial statistics**.

Worked Example: Undirected 2D Lattice for Image Modeling

Using an undirected 2D lattice to model the same 2D image removes the need for an arbitrary ordering:



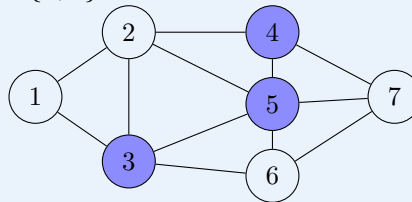
Now the **Markov blanket** of each node is simply its nearest neighbors (more on Markov blankets for UGMs below) – a much more natural notion of “local context” for spatial data than any DAG parent set.

5 Markov Properties of UGMs

Definition : Global Markov Property

Given sets of nodes A, B, C , $X_A \perp X_B \mid X_C$ if and only if C **separates** A from B in the graph $\mathcal{G}$, i.e. there are **no paths** connecting any node in A to any node in B once all nodes in C are removed.

Example: in the 7-node graph below, $\{X_1, X_2\} \perp \{X_6, X_7\} \mid \{X_3, X_4, X_5\}$, since removing $\{3, 4, 5\}$ disconnects $\{1, 2\}$ from $\{6, 7\}$.



Definition : Local Markov Property (Markov Blanket)

The set of nodes that renders a node X_s conditionally independent of all other nodes in $\mathcal{G}$:

$$X_s \perp \mathcal{V} \setminus \{\text{mb}(X_s), X_s\} \mid \text{mb}(X_s),$$

where $\text{mb}(X_s)$ is the **Markov blanket** of X_s – for a UGM this is simply the set of neighbors of X_s .

Example: in the graph above, $\text{mb}(X_1) = \{X_2, X_3\}$, i.e. $X_1 \perp \{X_4, X_5, X_6, X_7\} \mid \{X_2, X_3\}$.

Definition : Pairwise Markov Property

Two nodes X_s and X_t are conditionally independent given the rest if there is **no direct edge** between them:

$$X_s \perp X_t \mid \mathcal{V} \setminus \{X_s, X_t\}, \quad \text{where } \mathcal{E}_{st} = \emptyset.$$

Example: in the graph above, $X_1 \perp X_7 \mid \{X_2, X_3, X_4, X_5, X_6\}$.

Key Rule: Equivalence of the Markov Properties

$$G \implies L \implies P, \quad \text{and (assuming } p(x) > 0 \text{ for all } x) \quad P \implies G.$$

It is **obvious** that global Markov implies local Markov, which implies pairwise Markov. What is **less obvious**, but true whenever $p(x) > 0$ for all x , is that pairwise Markov also implies global Markov – so under positivity, all three properties are **equivalent**. (This last step relies on the **Intersection** rule of Section 2.3, which requires $p(x) > 0$; that is precisely where the positivity assumption enters here.)

Proof Sketch: Global $\Rightarrow$ Local

This is the special case of the global Markov property where $X_A = X_s$, $X_C = \text{mb}(X_s)$, and $X_B = \mathcal{V} \setminus \{\text{mb}(X_s), X_s\}$:

$$X_A \perp X_B \mid X_C \implies X_s \perp \mathcal{V} \setminus \{\text{mb}(X_s), X_s\} \mid \text{mb}(X_s).$$

■

Proof Sketch: Local $\Rightarrow$ Pairwise

Given any node X_t that is **not** adjacent to X_s , it follows from the local Markov property that

$$X_s \perp X_{\mathcal{V} \setminus \{\text{mb}(X_s), X_s\}} \mid \text{mb}(X_s).$$

This implies $X_s \perp X_t \mid X_{\mathcal{V} \setminus \{X_s, X_t\}}$, i.e. the pairwise Markov property. ■

Proof Sketch: Pairwise $\Rightarrow$ Global (assuming $p(x) > 0$)

Let $S, A, B, D \subset \mathcal{V}$ be disjoint sets of nodes with S separating A from B in $\mathcal{G}$, where $A \neq \emptyset$ and $B \neq \emptyset$. Let $d = |\mathcal{V}|$. We proceed by **backward induction** on $|S|$.

Base case ($|S| = d - 2$): then $|A| = |B| = 1$, so $A \perp B \mid S$ is exactly the pairwise Markov property.

Inductive case ($|S| < d - 2$): WLOG assume the (non-empty) set of remaining nodes $D = \mathcal{V} \setminus (A \cup B \cup S)$ is not connected to B . By the inductive hypothesis (applied to larger separating sets),

$$A \perp B \mid \{S, D\} \quad \text{and} \quad B \perp D \mid \{A, S\}.$$

Applying the **Intersection** rule (Section 2.3) – which needs $p(x) > 0$, exactly the assumption carried through this whole proof:

$$A \perp B \mid \{S, D\} \text{ and } B \perp D \mid \{A, S\} \implies B \perp \{A, D\} \mid S \implies \text{global Markov.}$$

■

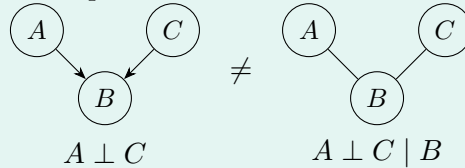
The importance of this result is that it is usually **easier to empirically assess pairwise conditional independence**; such pairwise CI statements can then be used to **construct a graph** from which global CI statements are extracted.

6 Comparative Semantics: DGMs vs. UGMs

We have seen that it is easier to determine conditional independence using UGMs than DGMs. A natural question: can we determine conditional independence in a DGM using a UGM, or vice versa? **This is not possible in general!**

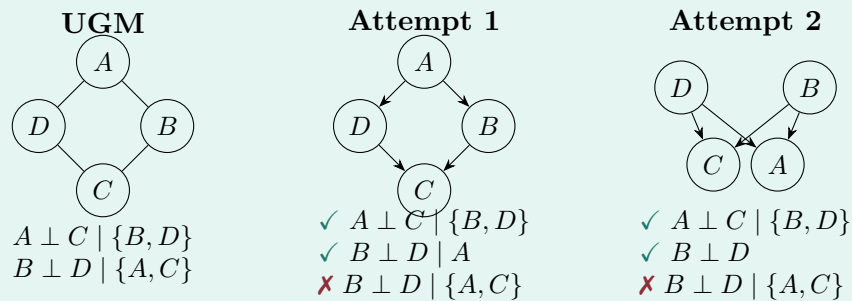
Worked Example: Dropping Edge Orientations Is Not Always Correct

It is tempting to simply convert a DGM to a UGM by dropping the orientation of the edges – but this can silently introduce independencies that are not actually present.



The DGM (a head-to-head / v-structure at B) encodes the marginal independence $A \perp C$, but **not** $A \perp C \mid B$. The naively-undirected graph on the right, however, **does** encode $A \perp C \mid B$ – this conditional independence is **not** present in the original DGM!

Worked Example: CI Relationships a UGM Can Model but a DGM Cannot



No DAG can simultaneously encode both $A \perp C \mid \{B, D\}$ and $B \perp D \mid \{A, C\}$ – every DAG attempt introduces an extra (unwanted) dependency or fails to break one required by the UGM. (Image source: Koller and Friedman, 2009.)

7 Parameterization of MRFs

It is difficult to do local parameterization based on conditional probabilities, since there is **no topological ordering** associated with UGMs. It turns out to be better to **abandon conditional probabilities** altogether and use some (non-probabilistic) functions instead – we **lose** the ability to give a local probabilistic interpretation to these functions, but we **retain** the all-important representation of the joint as a product of local functions.

How do we decide the domain of the local functions? Recall that two nodes X_i, X_j not directly linked in a UGM are conditionally independent given all other nodes. Thus it must be possible to obtain a factorization of the joint that places X_i and X_j in *different* factors – so we cannot have a local function that depends on both X_i and X_j unless they are linked, i.e. $p(x_1, \dots, x_N) \neq \psi_1(x_i, x_j, \dots) \dots \psi_m(\dots)$ when $\mathcal{E}_{ij} = \emptyset$.

Definition : Clique and Maximal Clique

A **clique** of a graph is a fully-connected subset of nodes. The **maximal cliques** of a graph are the cliques that cannot be extended to include additional nodes without losing the fully-connected property.

Conditional independence is impossible for any two nodes within a maximal clique! This suggests that all nodes X_C belonging to a maximal clique C should appear together in one local function $\psi(x_C)$.

Key Rule: Hammersley–Clifford Theorem

A positive distribution $p(\mathbf{y}) > 0$ satisfies the conditional-independence properties of an undirected graph $\mathcal{G}$ if and only if p can be represented as a **product of factors, one per maximal clique**:

$$p(\mathbf{y} \mid \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c \mid \boldsymbol{\theta}_c)$$

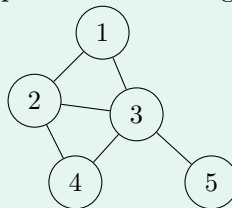
where $\mathcal{C}$ is the set of all maximal cliques of $\mathcal{G}$, $\psi_c(\cdot)$ is the **factor** (or **potential function**) of clique c , $\boldsymbol{\theta}_c$ is the parameter of $\psi_c(\cdot)$, and $Z(\boldsymbol{\theta})$ is the **partition function**:

$$Z(\boldsymbol{\theta}) \triangleq \sum_{\mathbf{y}} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c \mid \boldsymbol{\theta}_c).$$

Since $\psi_c(\cdot)$ can be any arbitrary **positive** function, the partition function $Z(\boldsymbol{\theta})$ ensures the overall distribution sums (or integrates) to 1.

Worked Example: Maximal-Clique Factorization

Assume $y_i \in \{0, 1\}$ for the 5-node graph below with edges 1–2, 1–3, 2–3, 2–4, 3–4, 3–5:



The maximal cliques are $\{1, 2, 3\}$, $\{2, 3, 4\}$ and $\{3, 5\}$, giving

$$p(\mathbf{y} \mid \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \psi_{123}(y_1, y_2, y_3) \psi_{234}(y_2, y_3, y_4) \psi_{35}(y_3, y_5),$$

$$Z(\boldsymbol{\theta}) = \sum_{\mathbf{y}} \psi_{123}(y_1, y_2, y_3) \psi_{234}(y_2, y_3, y_4) \psi_{35}(y_3, y_5).$$

(Image source: M. I. Jordan, “An Introduction to Probabilistic Graphical Models”, 2002.)

Key Rule: Pairwise Parameterization of MRFs

Parameterization of an MRF is **not unique**! We are free to relax the parameterization to the

edges of the graph rather than the maximal cliques:

$$p(\mathbf{y} \mid \boldsymbol{\theta}) \propto \psi_{12}(y_1, y_2) \psi_{13}(y_1, y_3) \psi_{23}(y_2, y_3) \psi_{24}(y_2, y_4) \psi_{34}(y_3, y_4) \psi_{35}(y_3, y_5) \propto \prod_{s \sim t} \psi_{st}(y_s, y_t).$$

This is the **pairwise MRF**, widely used for its simplicity, although it is not as general as the maximal-clique parameterization.

Worked Example: Canonical Parameterization of MRFs

Another way is to define the parameterization over **all** cliques in the graph (not just maximal ones), i.e. the **canonical parameterization**; a **uniform prior** can be assumed on any potential function we choose to drop (optional). For the same 5-node graph:

$$\begin{aligned} p(\mathbf{y} \mid \boldsymbol{\theta}) &\propto \psi_1(y_1) \psi_2(y_2) \psi_3(y_3) \psi_4(y_4) \psi_5(y_5) \cancel{\psi_{12} \psi_{23} \psi_{13} \psi_{24} \psi_{34} \psi_{35}} \psi_{123}(y_1, y_2, y_3) \psi_{234}(y_2, y_3, y_4) \\ &= \psi_1(y_1) \psi_2(y_2) \psi_3(y_3) \psi_4(y_4) \psi_5(y_5) \psi_{123}(y_1, y_2, y_3) \psi_{234}(y_2, y_3, y_4) \end{aligned}$$

i.e. once the maximal-clique potentials ψ_{123}, ψ_{234} (and ψ_{35}) are included, the lower-order potentials on their subsets can be set to a uniform prior (crossed out) without changing the represented distribution.

8 Gibbs Distributions and Log-Linear Potentials

Definition : Gibbs Distribution

There is a deep connection between UGMs and **statistical physics**. A **Gibbs distribution** can be written as

$$p(\mathbf{y} \mid \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp \left(- \sum_c E(\mathbf{y}_c \mid \boldsymbol{\theta}_c) \right),$$

where $E(\mathbf{y}_c) > 0$ is the **energy** associated with the variables in clique c .

Key Rule: Gibbs Distributions Are UGMs (Energy-Based Models)

We can convert a Gibbs distribution into a UGM by defining $\psi_c(\mathbf{y}_c \mid \boldsymbol{\theta}_c) = \exp(-E(\mathbf{y}_c \mid \boldsymbol{\theta}_c))$, so that

$$p(\mathbf{y} \mid \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp \left(- \sum_c E(\mathbf{y}_c \mid \boldsymbol{\theta}_c) \right) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c \mid \boldsymbol{\theta}_c).$$

High probability states correspond to low energy configurations. Such models are also known as **energy-based models**, which is where the term “potential function” for $\psi_c(\cdot)$ comes from.

Definition : Log-Linear Potential Functions

Potentials represent the relative “**compatibility**” between different assignments of the random variables. A general approach is to define the **log potentials** as a **linear function** of

parameters $\theta_c \in \mathbb{R}^M$:

$$-E(\mathbf{y}_c) = \log \psi_c(\mathbf{y}_c) \triangleq \phi_c(\mathbf{y}_c)^\top \theta_c,$$

where $\phi_c : y_c \mapsto \mathbb{R}^M$ is a **feature vector** derived from the values of the variables y_c . The resulting log probability has the form

$$\log p(\mathbf{y} \mid \theta) = \sum_c \phi_c(\mathbf{y}_c)^\top \theta_c - \log Z(\theta),$$

also known as a **maximum entropy** or **log-linear** model.

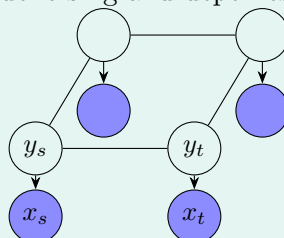
Worked Example: Log-Linear Features Reduce Parameter Count

For $y_c = \{y_1, y_2, y_3\}$ on the 5-node graph of Section 7, we set $M = |c| = 3$, $\theta_c \in \mathbb{R}^M$ and $\phi_c : y_c \mapsto [y_1, y_2, y_3]^\top$, giving $\psi_c(y_c) = \exp\{\phi_c(y_c)^\top \theta_c\}$. Instead of needing $2^3 = 8$ parameters to fully specify $\psi_{123}(y_1, y_2, y_3)$ over all binary assignments, we only need $\theta_c = [\theta_{c1}, \theta_{c2}, \theta_{c3}]$ – **3 parameters**.

9 Combining UGMs and DGMs

Worked Example: Ising and Potts Models

Combining unary (DGM-style, per-node) and pairwise (UGM-style) potentials is common in computer vision, e.g. binary image denoising and depth estimation from stereo:



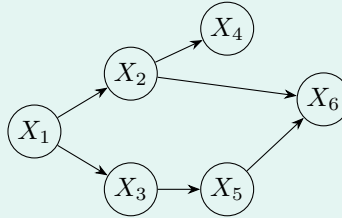
$$p(\mathbf{y}, \mathbf{x} \mid J, \theta) = p(\mathbf{y} \mid J) \prod_t p(x_t \mid y_t, \theta) = \underbrace{\left[\frac{1}{Z(J)} \prod_{s \sim t} \psi(y_s, y_t; J) \right]}_{\text{pairwise potential}} \underbrace{\prod_t p(x_t \mid y_t, \theta)}_{\text{unary potential}}.$$

Ising model: $y_i, x_i \in \{0, 1\}$, energy $E(y_s, y_t; J) = J |y_s - y_t|$, $J > 0$.

Potts model (multi-label generalization): $y_i, x_i \in \{1, \dots, L\}$, energy $E(y_s, y_t; J) = J \min(|y_s - y_t|, 1)$, $J > 0$.

Worked Example: DGM Conditionals as UGM Potentials

We can also use local conditional probabilities from a DGM to represent the potential functions of a UGM:

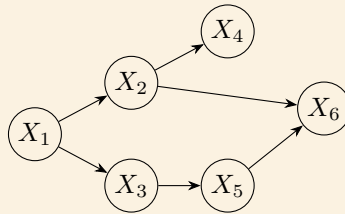


$$p(x) = \frac{1}{Z} \varphi_{12}(x_1, x_2) \varphi_{13}(x_1, x_3) \varphi_{24}(x_2, x_4) \varphi_{35}(x_3, x_5) \varphi_{256}(x_2, x_5, x_6)$$

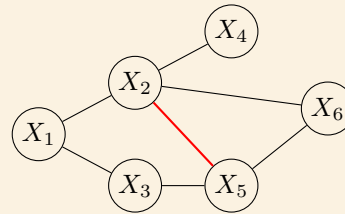
with $\varphi_{12} = p(x_2 | x_1)$, $\varphi_{13} = p(x_3 | x_1)$, $\varphi_{24} = p(x_4 | x_2)$, $\varphi_{35} = p(x_5 | x_3)$, $\varphi_{256} = p(x_6 | x_2, x_5)$, and (since the DGM is already normalized) $Z = 1/p(x_1)$.

Key Rule: Moralization

A DGM can be converted into a UGM by “**marrying**” the unmarried parents of each node – i.e. connecting co-parents with an undirected edge and dropping all arrowheads – called **moralization**.



$$x_2 \perp \{x_3, x_5\} \mid x_1$$



$$x_2 \not\perp \{x_3, x_5\} \mid x_1$$

This process **preserves the joint distribution**, but **conditional independence is lost** (the moralizing edge, shown in red, removes $x_2 \perp x_5 \mid x_1$ that held in the DGM). Moralization is important for exact inference, covered in later lectures.

10 Conditional Random Fields (CRFs)

Definition : Generative vs. Discriminative Models

- **Generative models:** approaches that explicitly or implicitly model the distribution of inputs and outputs; sampling from the distribution, it is possible to generate synthetic data points in the input space. **Likelihood:** $p(\mathbf{x} \mid \mathcal{C}_k)$.
- **Discriminative models:** approaches that model the **posterior** probabilities directly. **Posterior:** $p(\mathcal{C}_k \mid \mathbf{x})$.

Definition : Conditional Random Field (CRF)

A CRF, or **discriminative random field**, is a version of an MRF where all the clique potentials are **conditioned on all observed random variables $\mathbf{x}$** :

$$p(\mathbf{y} \mid \mathbf{x}, \mathbf{w}) = \frac{1}{Z(\mathbf{x}, \mathbf{w})} \prod_c \psi_c(\mathbf{y}_c \mid \mathbf{x}, \mathbf{w}),$$

usually with a **log-linear** representation of the potentials, $\psi_c(\mathbf{y}_c \mid \mathbf{x}, \mathbf{w}) = \exp(\mathbf{w}_c^\top \phi(\mathbf{x}, \mathbf{y}_c))$, where $\phi(y_c, \mathbf{x})$ is a feature vector derived from the latent variables Y_c and the observed variables

$\mathbf{x}$, and $\mathbf{w}_c$ are the parameters.

Key Rule: CRF vs. MRF – Advantages and Disadvantages

Advantages of CRFs:

- No need to “waste resources” modeling things we always observe.
- We can make the potentials (or factors) of the model **data-dependent** – e.g. in NLP, latent labels can depend on observations of the sentence, such as which language it is written in.
- Focuses attention on modeling what we care about: the distribution of latent given observed random variables.

Disadvantages of CRFs:

- **Not generative** – we cannot generate new samples.
- **Learning is slower** (more detail in coming lectures).

Worked Example: CRF as Logistic Regression

The Naïve Bayes classifier **ignores dependencies** among the observations $X_1, \dots, X_N$: $p(x_1, \dots, x_N, Y) = \prod_{n=1}^N p(x_n | Y)$, with prediction via Bayes' rule.

In a CRF, we instead **model the full dependencies** among $X_1, \dots, X_N$:

$$\begin{aligned} p(\mathbf{y} | \mathbf{x}, \mathbf{w}) &= \frac{1}{Z(\mathbf{x}, \mathbf{w})} \psi(\mathbf{y} | \mathbf{x}, \mathbf{w}) = \frac{\exp(\phi(\mathbf{y}, \mathbf{x})^\top \mathbf{w})}{\sum_{\mathbf{y}} \exp(\phi(\mathbf{y}, \mathbf{x})^\top \mathbf{w})} \\ &= \frac{\exp(\sum_n \phi_n(\mathbf{y}, \mathbf{x}) w_n)}{\sum_{\mathbf{y}} \exp(\sum_n \phi_n(\mathbf{y}, \mathbf{x}) w_n)} \quad (\text{log-linear model}). \end{aligned}$$

For $y \in \{0, 1\}$ (e.g. spam classification), define $\phi_n(y=1, \mathbf{x}) w_n = \mathbb{I}(y^+) x_n w_n^+$ and $\phi_n(y=0, \mathbf{x}) w_n = \mathbb{I}(y^-) x_n w_n^-$. Then

$$p(y^+ | \mathbf{x}) = \frac{\exp(\mathbf{w}^{+\top} \mathbf{x})}{\exp(\mathbf{w}^{+\top} \mathbf{x}) + \exp(\mathbf{w}^{-\top} \mathbf{x})} = \frac{1}{1 + \exp(-\mathbf{w}'^\top \mathbf{x})}, \quad \mathbf{w}' = \mathbf{w}^+ - \mathbf{w}^-,$$

which is exactly **logistic regression**! Note there is really just one set of parameters $\mathbf{w}'$ – we don't need to separately model $\mathbf{w}^+$ and $\mathbf{w}^-$. The CRF becomes a **deep network** once $\mathbf{w}'^\top \mathbf{x}$ is replaced by a nonlinear function $f(\mathbf{x}; \mathbf{w})$: $p(y^+ | \mathbf{x}) = \sigma(f(\mathbf{x}; \mathbf{w}))$, where $\sigma(\cdot)$ is the sigmoid (or softmax, for multiclass classification).

Worked Example: Linear-Chain CRF vs. HMM vs. Chain MRF

For sequential data, compare three models with the same graph topology:

$$\text{HMM (generative): } p(\mathbf{x}, \mathbf{y} | \mathbf{w}) = \prod_{t=1}^T p(y_t | y_{t-1}, \mathbf{w}) p(\mathbf{x}_t | y_t, \mathbf{w})$$

$$\text{Chain MRF (generative): } p(\mathbf{x}, \mathbf{y} | \mathbf{w}) = \prod_{t=1}^T \psi(y_t; x_t, \mathbf{w}) \prod_{t=1}^{T-1} \psi(y_t, y_{t+1}; \mathbf{w})$$

$$\text{Chain CRF (discriminative): } p(\mathbf{y} \mid \mathbf{x}, \mathbf{w}) = \frac{1}{Z(\mathbf{x}, \mathbf{w})} \prod_{t=1}^T \psi(y_t; \mathbf{x}, \mathbf{w}) \prod_{t=1}^{T-1} \psi(y_t, y_{t+1}; \mathbf{x}, \mathbf{w})$$

Label bias problem: in the HMM and chain MRF, local features at time t do not influence states prior to t – X_t is **d-separated** from all other nodes at Y_t (given Y_t 's neighbors), blocking information flow backward through the chain. In the chain CRF, every Y_t connects to the *whole* observation sequence $\mathbf{x}$, so this problem is alleviated.

Example (POS tagging): the word “banks” could be a verb (“he banks at DBS”) or a noun (“the river banks were overflowing”) – locally the tag is ambiguous. If we later see the word “fishing”, that gives enough context to infer “river banks”. In an HMM or chain MRF this future evidence is d-separated from the earlier tag; a chain CRF, which conditions every position on the entire observed sequence, does not suffer from this limitation. (Image source: E. Xing, CMU 10-708, Lecture 3.)

11 Applications

- **Multi-target tracking:** an MRF is used to model interactions between ants in a video, so that every ant can be tracked jointly rather than independently (Zia Khan, Tucker Balch, Frank Dellaert, “An MCMC-based Particle Filter for Tracking Multiple Interacting Targets”, ECCV 2014).
- **Road-network extraction:** a CRF is used to infer the location of road-segment centerlines together with their width from aerial images (G. Mattyus, S. Wang, S. Fidler, R. Urtasun, “Enhancing World Maps by Parsing Aerial Images”, ICCV 2015).

12 Summary

- Derived the five rules of conditional independence: Symmetry, Decomposition, Weak Union, Contraction, Intersection.
- Used I-Map to explain the correctness of graphical models for conditional independence.
- Explained the Markov properties (global, local, pairwise) of a UGM and used them to find all conditional independencies.
- Used clique potential functions to parameterize an MRF – representing the joint distribution as a product of clique potentials (Hammersley–Clifford theorem).
- Described the differences and similarities between a Markov Random Field and a Conditional Random Field.

Key Takeaways

1. **Five rules of conditional independence** (Symmetry, Decomposition, Weak Union, Contraction, Intersection) let us manipulate CI statements algebraically. Symmetry is a trivial biconditional; Weak Union and Contraction hold **unconditionally**, and Contraction’s converse (which follows from Weak Union + Decomposition) does too, as does Intersection’s converse (which follows from Weak Union alone). **Intersection’s forward direction is the sole exception:** it is the converse of Weak Union but is only valid for **strictly positive** distributions ($p(x) > 0$) – without positivity it can fail, so “pairwise CI $\Rightarrow$ joint CI” is not guaranteed in general. Decomposition’s converse is a second, unrelated failure: it is false in general regardless of positivity (XOR counterexample). The

Intersection rule is also the key tool linking pairwise to global Markov, in a step that needs $p(x) > 0$.

2. A graph $\mathcal{G}$ is a valid **I-Map** of p iff $I(\mathcal{G}) \subseteq I(p)$; this criterion lets us go from graphs to factorizations of the joint distribution.
3. **UGMs (MRFs)** drop edge orientation, avoiding the awkward, arbitrary choices DGMs force on domains like image analysis; but DGMs and UGMs can each encode CI relationships the other cannot.
4. The **global, local, and pairwise Markov properties** are equivalent whenever $p(x) > 0$, letting us read off (or empirically build) conditional independencies directly from the graph structure or from pairwise tests.
5. The **Hammersley–Clifford theorem** parameterizes an MRF as a normalized product of potential functions over maximal cliques; this connects to **Gibbs distributions** (low energy $\leftrightarrow$ high probability) and **log-linear** potentials, which compactly parameterize cliques via feature vectors.
6. **CRFs** condition all clique potentials on the observed variables, trading away the generative property for data-dependent potentials and (in chain models) freedom from the label-bias / d-separation problem that limits HMMs and chain MRFs.

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